Column Space and Nullspace

Every matrix carries two spaces. The column space answers "which targets can I reach?". The nullspace answers "which trades change nothing?". Together they describe every answer to $Ax = b$.

PART 1 Index

What this lesson covers, in order. Click a topic to jump to it.

1. **When can Ax = b be solved?**
   Exactly when b is in the column space. MIT's 4 × 3 example and our funds with a balanced fund.
2. **The nullspace: trades that change nothing**
   All x with $Ax = 0$. Found by elimination on a 2 × 2 and a 3 × 3: the trade $(1, 1, -2)$.
3. **Why the nullspace is a subspace**
   Two do-nothing trades add to a do-nothing trade. Column space vs nullspace side by side.
4. **When the nullspace is only zero**
   With midcap, elimination leaves no free fund: every target has exactly one portfolio.
5. **All solutions: one answer + the nullspace**
   Every portfolio that hits a target, as a slider. Why these solutions are not a subspace.
6. **Spotting subspaces: rules that equal zero**
   Zero-cost and market-neutral trades are nullspaces; meeting both is an intersection.

METHODS IN THIS LESSON: Ax = b solvable ⇔ b ∈ C(A) · Nullspace N(A) · Elimination on Ax = 0 · N(A) is a subspace · N(A) = {0} test · Particular + nullspace · Intersection of nullspaces

PART 2 Prerequisites

Click any link to review it, then come back.

- The column space C(A) (Lesson 5): the targets your funds can reach, and the plane $R = 7B + 5E$.
- Vector spaces and subspaces (Lesson 5): closed under adding and scaling; zero-cost trades.
- Elimination, 3 × 3 and back substitution (Lesson 2).
- Singular matrices (Lesson 1): the balanced fund that added nothing new.

PART 3

Explanation

THE WHOLE IDEA IN ONE LINE

Column space: which targets can my funds reach? **Nullspace:** which trades change nothing at all? Every portfolio that hits a target = **one** portfolio that hits it + **any** do-nothing trade.

THE RUNNING EXAMPLE: THE SAME FUNDS, ONE NEW ONE

Rules (rows): budget (₹ lakh), return (₹ thousand), market exposure. Each fund is a column: what ₹1 lakh of it does to the three rules.

Fund	Return	Beta	Column (budget, return, exposure)
Nifty	12%	1.0	$(1, 12, 1)$
Bonds	7%	0	$(1, 7, 0)$
Midcap	15%	1.2	$(1, 15, 1.2)$
Balanced (half Nifty, half bonds)	9.5%	0.5	$(1, 9.5, 0.5)$

We'll use two fund matrices: one with the **balanced** fund (it repeats what Nifty and bonds already do) and one with **midcap** (it adds something new).

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 12 & 7 & 9.5 \\ 1 & 0 & 0.5 \end{bmatrix} \text{ (Nifty, bonds, balanced)} \qquad A_{\text{mid}} = \begin{bmatrix} 1 & 1 & 1 \\ 12 & 7 & 15 \\ 1 & 0 & 1.2 \end{bmatrix} \text{ (Nifty, bond}$$

1. When can Ax = b be solved?

THE WHOLE IDEA IN ONE LINE

Ax is always a mix of the columns of A. So $Ax = b$ has an answer **exactly when b is a mix of the columns**, that is, when b is in the column space $C(A)$.

Step 1: Ax is a mix of the columns

Hold ₹7 lakh Nifty, ₹3 lakh bonds, nothing in balanced: $x = (7, 3, 0)$. Row by row:

$$Ax = \begin{bmatrix} 1 & 1 & 1 \\ 12 & 7 & 9.5 \\ 1 & 0 & 0.5 \end{bmatrix} \begin{bmatrix} 7 \\ 3 \\ 0 \end{bmatrix} = \begin{bmatrix} 1(7)+1(3)+1(0) \\ 12(7)+7(3)+9.5(0) \\ 1(7)+0(3)+0.5(0) \end{bmatrix} = \begin{bmatrix} 7+3+0 \\ 84+21+0 \\ 7+0+0 \end{bmatrix} = \begin{bmatrix} 10 \\ 105 \\ 7 \end{bmatrix}.$$

Column by column, the same product is $7\times$ (Nifty column) $+ 3\times$ (bonds column) $+ 0\times$ (balanced column). Whatever x you pick, Ax is some mix of the columns. That's all a matrix can produce.

Step 2: so the target must be a mix too

> **RULE · AX = B IS SOLVABLE EXACTLY WHEN B IS IN C(A)**
>
> If $Ax = b$ has a solution, then $b = Ax$ is a mix of the columns, so $b \in C(A)$. If $b \in C(A)$, the amounts of each column in that mix **are** a solution x. In money terms: a target can be hit exactly when it's a target your funds can reach.

A handy way to **build** reachable targets: pick any x first, then compute $b = Ax$. Every b you make this way is in $C(A)$, and you already know one answer for it.

Step 3: MIT's example: 4 rules, 3 unknowns

Strang's matrix has four rows (four equations) and only three columns (three unknowns):

$$A_{\text{MIT}} = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 1 & 3 \\ 3 & 1 & 4 \\ 4 & 1 & 5 \end{bmatrix}.$$

1 WHAT — Its columns live in $\mathbb{R}^4$ (four entries each). Can three columns reach every target in $\mathbb{R}^4$?

WHY NOT — Three directions can fill at most a 3D slice of a 4D space. Some targets must be out of reach.

2 LOOK CLOSER — Column 3 = column 1 + column 2: $1+1=2, 2+1=3, 3+1=4, 4+1=5$ ✓.

SO — Column 3 adds no new direction. $C(A_{\text{MIT}})$ is only the **plane** of mixes $a(1,2,3,4) + c(1,1,1,1) = (a+c,\ 2a+c,\ 3a+c,\ 4a+c)$, a 2D plane inside $\mathbb{R}^4$.

3	REACHABLE	Each entry is a more than the one before, so the four entries must go up in **equal steps**. $(1, 2, 3, 4)$: steps 1, 1, 1 ✓. $(2, 3, 4, 5)$ ✓.
	NOT REACHABLE	$(1, 0, 0, 0)$: steps $-1, 0, 0$, not equal ✗. No x solves $A_{\text{MIT}}x = (1, 0, 0, 0)$.

Step 4: the same thing with our funds

The balanced fund is half Nifty, half bonds: $\frac{1}{2}(1, 12, 1) + \frac{1}{2}(1, 7, 0) = (0.5 + 0.5,\ 6 + 3.5,\ 0.5 + 0) = (1, 9.5, 0.5)$ ✓. Like column 3 above, it adds no new direction. So $C(A)$ is still the plane from Lesson 5: targets with $R = 7B + 5E$. The target $(10, 105, 7)$ passes ($70 + 35 = 105$); $(10, 105, 5)$ still fails ($70 + 25 = 95$).

IN ONE LINE

A target is hittable exactly when it's in $C(A)$; a column that's a mix of the others makes $C(A)$ no bigger.

2. The nullspace: trades that change nothing

THE WHOLE IDEA IN ONE LINE

The **nullspace** $N(A)$ is every trade x with $Ax = 0$: trades that move **no** rule at all. Same budget, same return, same exposure, just different funds.

METHOD · NULLSPACE N(A)

For a matrix A with n columns, $N(A)$ is the set of all x in $\mathbb{R}^n$ with $Ax = 0$. It always contains $x = 0$ ("do nothing"). The question is whether it contains anything else.

Step 1: two twin index funds (2 × 2)

Two Nifty index funds from different fund houses do exactly the same thing. With just the budget and return rules, both columns are $(1, 12)$:

$$A_{\text{twin}} = \begin{bmatrix} 1 & 1 \\ 12 & 12 \end{bmatrix}, \qquad \text{solve } A_{\text{twin}} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

METHOD · ELIMINATION ON AX = 0

Eliminate as usual (the right side is all zeros and stays zeros, so we leave it out). A column with no pivot gives a **free** unknown: choose it, then back-substitute for the rest. No free unknown means the only answer is $x = 0$.

1

WHAT WE DO Clear x out of row 2.

HOW Multiplier $= 12/1 = 12$. Row 2 $-$ $12\times$ row 1: $12 - 12(1) = 0$, $12 - 12(1) = 0$.

NOW Row 2 is $0x + 0y = 0$: always true, no information. Column 2 has no pivot, so y is **free**.

2

CHOOSE $y = -1$. Row 1: $x + (-1) = 0$, so $x = 1$.

CHECK Budget: $1(1) + 1(-1) = 0$ ✓. Return: $12(1) + 12(-1) = 0$ ✓.

MEANING $N(A_{\text{twin}})$ = all multiples $c(1, -1)$: move ₹c lakh from one twin to the other. Nothing changes. A **line** of do-nothing trades.

Compare Nifty and bonds, $\begin{bmatrix} 1 & 1 \\ 12 & 7 \end{bmatrix}$: row 2 $-$ $12\times$ row 1 gives $12 - 12 = 0$ and $7 - 12 = -5$, so row 2 reads $-5y = 0$. Then $y = 0$ and row 1 gives $x = 0$. Only the zero trade: $N = \{0\}$. Different funds, so nothing can be swapped for free.

Step 2: Nifty, bonds and balanced (3 × 3)

Solve $Ax = 0$ for A = (Nifty, bonds, balanced), one entry at a time.

1

WHAT WE DO Clear Nifty out of the return row (row 2).

WHY The pivot is the 1 in row 1, column 1.

HOW Multiplier $= 12/1 = 12$. Row 2 $-$ $12\times$ row 1: $12 - 12(1) = 0$, $7 - 12(1) = -5$, $9.5 - 12(1) = -2.5$.

NOW Row 2 $= (0, -5, -2.5)$.

2

WHAT WE DO Clear Nifty out of the exposure row (row 3).

HOW Multiplier $= 1/1 = 1$. Row 3 $-$ $1\times$ row 1: $1 - 1 = 0$, $0 - 1 = -1$, $0.5 - 1 = -0.5$.

NOW Row 3 $= (0, -1, -0.5)$.

3 WHAT WE DO — Clear bonds out of row 3, using the pivot -5 in row 2.

HOW — Multiplier $= (-1)/(-5) = 0.2$. Row 3 $-$ $0.2\times$ row 2: $0 - 0.2(0) = 0,\ -1 - 0.2(-5) = -1 + 1 = 0,\ -0.5 - 0.2(-2.5) = -0.5 + 0.5 = 0.$

NOW — Row 3 is all zeros. Column 3 (balanced) has **no pivot**, so the balanced amount z is free.

$$U = \begin{bmatrix} \boxed{1} & 1 & 1 \\ 0 & \boxed{-5} & -2.5 \\ 0 & 0 & 0 \end{bmatrix}$$ pivots in columns 1 and 2; column 3 is free.

1 CHOOSE — $z = -2$ (any number works; -2 keeps the others whole).

ROW 2 — $-5y - 2.5(-2) = 0 \Rightarrow -5y + 5 = 0 \Rightarrow y = 1.$

ROW 1 — $x + 1 + (-2) = 0 \Rightarrow x = 1.$

2 CHECK — Budget: $1(1) + 1(1) + 1(-2) = 1 + 1 - 2 = 0$ ✓. Return: $12(1) + 7(1) + 9.5(-2) = 12 + 7 - 19 = 0$ ✓. Exposure: $1(1) + 0(1) + 0.5(-2) = 1 + 0 - 1 = 0$ ✓.

MEANING — Buy ₹1 lakh Nifty and ₹1 lakh bonds, sell ₹2 lakh balanced. You have just built the balanced fund yourself, so **nothing changes**.

Every multiple works too: $c(1, 1, -2)$. So $N(A)$ is a **line through the origin** in $\mathbb{R}^3$, the space of trades. You'll see it drawn in topic 5.

> **IN ONE LINE**
>
> The nullspace collects the do-nothing trades; a free column in elimination is a sign it holds more than zero.

3. Why the nullspace is a subspace

Run the subspace test from Lesson 5 on "all trades that change nothing".

> **RULE · N(A) IS ALWAYS A SUBSPACE**
>
> If $Av = 0$ and $Aw = 0$, then $A(v + w) = Av + Aw = 0 + 0 = 0$ and $A(cv) = c\,Av = c \cdot 0 = 0$. And $A0 = 0$. So sums, multiples and zero all stay inside.

1 ZERO — Doing nothing changes nothing: $A0 = 0$ ✓.

SUMS — Do one do-nothing trade, then another: still nothing changed. $(1, 1, -2) + (2, 2, -4) = (3, 3, -6)$, and $A(3, 3, -6) = (3 + 3 - 6,\ 36 + 21 - 57,\ 3 + 0 - 3) = (0, 0, 0)$ ✓.

MULTIPLES — Do it 5 times bigger, or in reverse: $-5 \times (1, 1, -2) = (-5, -5, 10)$, still 0 ✓.

Notice this works for **every** matrix, without solving anything. That's why Strang calls the nullspace and the column space the two subspaces every matrix comes with. They live in different places:

	Column space $C(A)$	**Nullspace** $N(A)$
Made of	targets b (outputs)	trades x (inputs)
Lives in	$\mathbb{R}^m$: one entry per **rule** (row)	$\mathbb{R}^n$: one entry per **fund** (column)
Question	Which targets can I reach?	Which trades change nothing?
How you get it	All mixes of the columns	All solutions of $Ax = 0$
Nifty, bonds, balanced	the plane $R = 7B + 5E$	the line through $(1, 1, -2)$

For MIT's 4×3 matrix the two spaces aren't even the same size of vector: $C(A_{\text{MIT}})$ sits in $\mathbb{R}^4$, and $N(A_{\text{MIT}})$ sits in $\mathbb{R}^3$. Its nullspace is the line through $(1, 1, -1)$ (column 1 + column 2 − column 3 $= 0$): row by row, $1 + 1 - 2 = 0, 2 + 1 - 3 = 0, 3 + 1 - 4 = 0, 4 + 1 - 5 = 0$ ✓.

> **IN ONE LINE**
> Do-nothing trades add and scale into do-nothing trades, so $N(A)$ is always a subspace, of the **trade** space $\mathbb{R}^n$.

4. When the nullspace is only zero

Swap the balanced fund for midcap. Does anything still change nothing?

1 ROW 2 — Multiplier 12. Row 2 $-$ $12\times$ row 1: $12 - 12 = 0,\ 7 - 12 = -5,\ 15 - 12 = 3$. Now $(0, -5, 3)$.

ROW 3

Multiplier 1. Row 3 − row 1: $1-1=0,\ 0-1=-1,\ 1.2-1=0.2$. Now $(0,-1,0.2)$.

2 ROW 3 AGAIN

Multiplier $(-1)/(-5)=0.2$. Row $3-0.2\times$ row 2: $0,\ -1-0.2(-5)=0,\ 0.2-0.2(3)=0.2-0.6=-0.4$.

NOW

$U=\begin{bmatrix}1&1&1\\0&-5&3\\0&0&-0.4\end{bmatrix}$: three pivots, **no free column**.

3 BACK-SUBSTITUTE

Row 3: $-0.4z=0\Rightarrow z=0$. Row 2: $-5y+3(0)=0\Rightarrow y=0$. Row 1: $x+0+0=0\Rightarrow x=0$.

MEANING

$N(A_{\text{mid}})=\{0\}$. No trade among these three funds leaves all three rules untouched.

RULE · N(A) = {0} MEANS NO COLUMN IS REDUNDANT

A non-zero x in $N(A)$ says "this mix of columns adds to zero", so one column can be rebuilt from the others. $N(A)=\{0\}$ says no column can be rebuilt: each adds a new direction. (Lesson 9 names this **independence**.)

Why a trader cares. When $N(A)=\{0\}$, each reachable target has **exactly one** portfolio: if $Ax=b$ and $Ay=b$, then $A(x-y)=0$, so $x-y=0$. That's why every client in Lessons 1–4 got a single answer. With the balanced fund, there's never just one answer, as the next topic shows.

▼ For the curious: midcap breaks the link, and a free trade appears

Midcap doesn't obey $R=7B+5E$ ($7+6=13\neq 15$). Take the trade $(-6,1,5)$ from Lesson 5: $A_{\text{mid}}(-6,1,5)=(-6+1+5,\ -72+7+75,\ -6+0+6)=(0,10,0)$. It changes only the return. That's a direction the balanced fund could never give, and it's why $C(A_{\text{mid}})$ is all of $\mathbb{R}^3$.

IN ONE LINE

No free column $\Rightarrow N(A)=\{0\}\Rightarrow$ every reachable target has one, and only one, portfolio.

5. All solutions: one answer + the nullspace

> **THE WHOLE IDEA IN ONE LINE**
>
> Find **one** portfolio that hits the target. Add any do-nothing trade and it still hits it. That gives **all** of them.

Step 1: one answer

Target $b = (10, 105, 7)$ with Nifty, bonds, balanced. From topic 1 we already know one answer: $x_p = (7, 3, 0)$ (the "p" is for **particular**).

Step 2: add the do-nothing trade

> **RULE · ALL SOLUTIONS = PARTICULAR + NULLSPACE**
>
> If $Ax_p = b$ and $Ax_n = 0$, then $A(x_p + x_n) = b + 0 = b$. And every solution comes this way: if $Ax = b$, then $A(x - x_p) = b - b = 0$, so $x - x_p$ is in $N(A)$. Here: $x = (7, 3, 0) + t(1, 1, -2) = (7 + t,\ 3 + t,\ -2t)$.

t	Portfolio (Nifty, bonds, balanced)	Budget	Return	Exposure
0	$(7, 3, 0)$	$7 + 3 + 0 = 10$	$84 + 21 + 0 = 105$	$7 + 0 + 0 = 7$
1	$(8, 4, -2)$	$8 + 4 - 2 = 10$	$96 + 28 - 19 = 105$	$8 + 0 - 1 = 7$
-1	$(6, 2, 2)$	$6 + 2 + 2 = 10$	$72 + 14 + 19 = 105$	$6 + 0 + 1 = 7$
-3	$(4, 0, 6)$	$4 + 0 + 6 = 10$	$48 + 0 + 57 = 105$	$4 + 0 + 3 = 7$

Four different portfolios, all with ₹10 lakh, ₹105 thousand and exposure 7. There are infinitely many: one for every t.

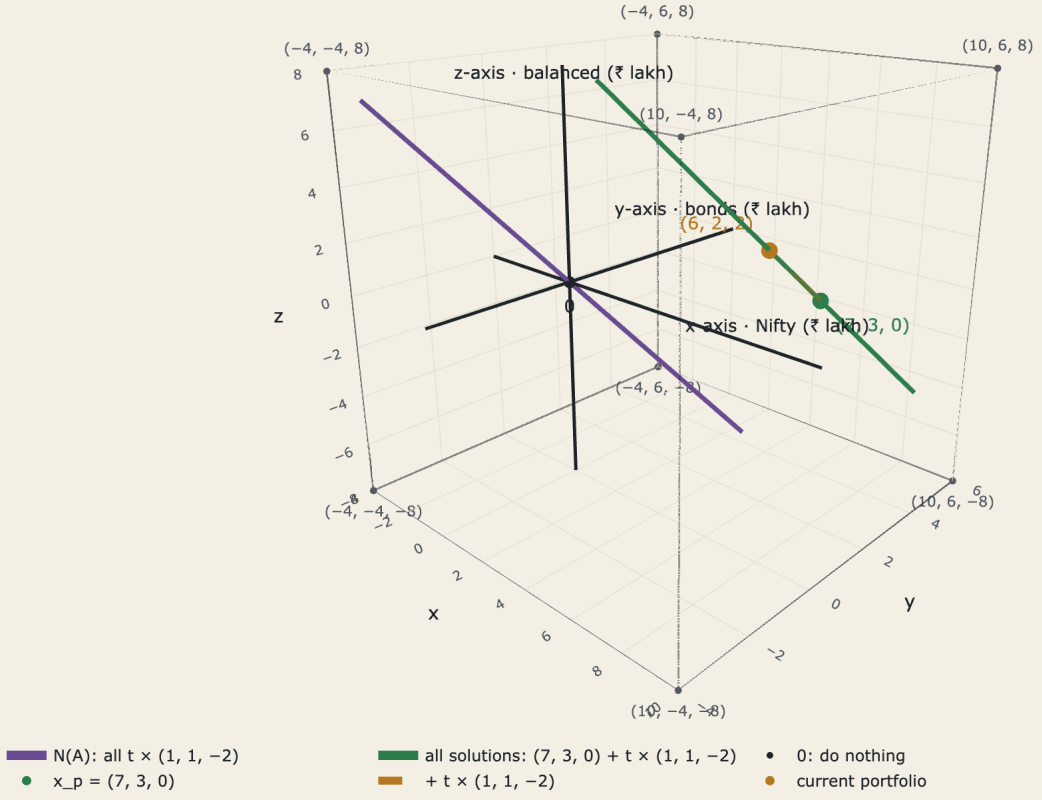
Try it: slide along the solutions

t = -1

Portfolio $= (7, 3, 0) + (-1) \times (1, 1, -2) = (6, 2, 2)$: ₹6 lakh Nifty, ₹2 lakh bonds, ₹2 lakh balanced.

Budget: $1(6) + 1(2) + 1(2) = 10$ ✓
Return: $12(6) + 7(2) + 9.5(2) = 105$ ✓
Exposure: $1(6) + 0(2) + 0.5(2) = 7$ ✓

Hits (10, 105, 7), and it is long-only (no negative amounts).



Two parallel lines in the space of trades (drag to rotate). Purple: the nullspace, do-nothing trades through 0. Green: every portfolio that hits $(10, 105, 7)$, the same line shifted to pass through $(7, 3, 0)$. The orange arrow is $t \times (1, 1, -2)$.

Step 3: these solutions are not a subspace

1 ZERO TEST

Is $(0, 0, 0)$ a solution? It would need $7 + t = 0$ and $3 + t = 0$ at once: $t = -7$ and $t = -3$. Impossible ✗. (Also, doing nothing has budget 0, not 10.)

SUM TEST

$(7, 3, 0) + (6, 2, 2) = (13, 5, 2)$. Budget $13 + 5 + 2 = 20$, return $156 + 35 + 19 = 210$, exposure $13 + 0 + 1 = 14$: that's $2b$, not b ✗.

PICTURE

The green line is the purple line **shifted off the origin.** A shifted line misses 0, like the fixed-budget line in Lesson 5.

MIT's version: $A_{\text{MIT}}x = (1, 2, 3, 4)$ is solved by $(1, 0, 0)$, and by $(1, 0, 0) + c(1, 1, -1)$ for every c. At $c = -1$ that's $(0, -1, 1)$: $-1(1, 1, 1, 1) + 1(2, 3, 4, 5) = (1, 2, 3, 4)$ ✓. A line through $(1, 0, 0)$ and $(0, -1, 1)$, but not through 0.

▼ For the curious: which of these portfolios are long-only?

Long-only needs every amount ≥ 0: $7 + t \geq 0$, $3 + t \geq 0$ and $-2t \geq 0$. The last says $t \leq 0$; the second says $t \geq -3$. So only $-3 \leq t \leq 0$ works: from $(7, 3, 0)$ (no balanced) to $(4, 0, 6)$ (no bonds). The algebra gives a whole line; the "no shorting" rule cuts it to a segment. That cut is what portfolio optimisers work with.

IN ONE LINE

All answers = one answer + the nullspace: a copy of $N(A)$ shifted to pass through x_p. A unique answer happens exactly when $N(A) = \{0\}$.

6. Spotting subspaces: rules that equal zero

THE WHOLE IDEA IN ONE LINE

Any set described by **linear rules that equal zero** is a nullspace, so it is automatically a subspace. Change one zero to anything else and it stops being one.

Step 1: zero-cost trades are a nullspace

With Nifty, bonds, midcap, a zero-cost trade has $x + y + z = 0$. That is $\begin{bmatrix} 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$: the nullspace of the 1×3 matrix $\begin{bmatrix} 1 & 1 & 1 \end{bmatrix}$. One rule, three unknowns, two free: a **plane** through 0.

Step 2: market-neutral trades are a nullspace

A trade that adds no market exposure has $1x + 0y + 1.2z = 0$: the nullspace of $\begin{bmatrix} 1 & 0 & 1.2 \end{bmatrix}$. Another plane through 0.

Step 3: both at once: stack the rules

RULE · STACKING RULES INTERSECTS NULLSPACES (MIT 6.3)

If $C = \begin{bmatrix} A \\ B \end{bmatrix}$ (the rows of A on top of the rows of B), then $Cx = 0$ means $Ax = 0$ **and** $Bx = 0$. So $N(C) = N(A) \cap N(B)$, and the intersection of two subspaces is a subspace.

$$C = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1.2 \end{bmatrix} \begin{matrix} \leftarrow \text{zero cost} \\ \leftarrow \text{zero exposure} \end{matrix}$$

1 ELIMINATE

Multiplier 1. Row 2 $-$ row 1: $1-1=0,\ 0-1=-1,\ 1.2-1=0.2$. Now $\begin{bmatrix} 1 & 1 & 1 \\ 0 & -1 & 0.2 \end{bmatrix}$: column 3 is free.

CHOOSE

$z = 5$. Row 2: $-y + 0.2(5) = 0 \Rightarrow y = 1$. Row 1: $x + 1 + 5 = 0 \Rightarrow x = -6$.

2 CHECK

Cost: $-6 + 1 + 5 = 0$ ✓. Exposure: $-6 + 0 + 1.2(5) = -6 + 6 = 0$ ✓.

MEANING

$N(C)$ is the line through $(-6, 1, 5)$: the trade from Lesson 5, now found as a nullspace. Two planes through 0 meet in a line through 0.

Step 4: what breaks it

Set of trades	Rule	Subspace?	Why
Zero-cost	$x + y + z = 0$	yes	linear, $= 0$: a nullspace
Zero-cost and market-neutral	both rules $= 0$	yes	intersection of two nullspaces
Budget exactly 10	$x + y + z = 10$	no	0 is not in it: a shifted plane
Zero-cost **or** market-neutral	either rule	no	a union: adding one of each can satisfy neither
Bonds = Nifty × midcap	$y = xz$	no	not linear: $(1,1,1)$ is in, $2\times$ it, $(2,2,2)$, is not $(2 \neq 4)$

IN ONE LINE

Linear rules set to zero always carve out a subspace (a nullspace); a non-zero right side, an "or", or a product breaks it.

▼ **Check yourself (MIT recitation): which of these sets of (b_1, b_2, b_3) are subspaces of $\mathbb{R}^3$? (1) $b_1 + b_2 - b_3 = 0$; (2) $b_3 = b_1 b_2$; (3) $(1,0,0) + c_1(1,0,-1) + c_2(1,0,1)$; (4) $(0,1,0) + c_1(1,0,-1) + c_2(1,0,1)$.**

(1) Yes: it's the nullspace of $\begin{bmatrix} 1 & 1 & -1 \end{bmatrix}$. (2) No: $(1,1,1)$ is in it ($1 = 1 \times 1$), but $(2,2,2)$ isn't ($2 \neq 2 \times 2$). (3) Yes: $(1,0,0) = \frac{1}{2}(1,0,-1) + \frac{1}{2}(1,0,1)$ already lies in the plane, so shifting by it changes nothing; the set is just the plane of all $c_1(1,0,-1) + c_2(1,0,1)$. (4) No: every mix of the two vectors has middle entry 0, so adding $(0,1,0)$ makes the middle entry 1. The zero vector is never reached.